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The flip side of destructive wildfires: a water supply buffer to climate change

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Keywords: Basin Characterization Model (BCM), Feather River watershed, wildfire, streamflow, soil moisture, evapotranspiration

29 Abstract:

30 Wet to dry climate extremes in California have directly or indirectly impacted droughts, wildfires, and floods
31 resulting in costly damage. Reservoirs in California were built to store snowpack into the summer and
32 provide multiple benefits in the state: they offer municipal and agricultural water resources during the dry
33 season and prevent flooding downstream. The upper Feather River watershed, which drains 9,300 square
34 kilometers (km²) into Lake Oroville, provides water to over 27 million people and 3,000 km² of farmland.
35 The North Complex fire in 2020 and Dixie and Sugar fires in 2021 burned over 50% of the watershed and
36 their long-term impact on the upper Feather River water supply was previously unknown. This study
37 quantified post-wildfire (postfire) changes in plant water use, soil moisture, and streamflow due to changes
38 in vegetation using a physically-based water balance model. Long term trends revealed decreases in
39 annual precipitation and mid to high elevation snowpack in this watershed from 1981-2020. Due to changes
40 in vegetation, annual postfire streamflow at Feather River at Oroville increased by 22%, 32% and 16%
41 while summer baseflows increased by 5%, 12% and 2% in water years 2022 through 2024, respectively.
42 Larger increases in streamflow occurred during water year 2023, an extreme wet year relative to the other
43 years, highlighting the challenges in quantifying and managing hydrologic systems with naturally high
44 interannual variability. Seasonal soil moisture increased by up to 34% depending on the location,
45 predominantly in fall and summer postfire. Despite observed historical decreases in precipitation and
46 snowpack, changes in vegetation due to wildfires or other disturbances may increase water availability, and
47 continued management of mixed-conifer forests (i.e. thinning and controlled burns) may help offset future
48 reduced precipitation and snowpack and therefore increase water supply resilience.

1. INTRODUCTION

California's extreme weather phenomena —cycles of floods, droughts, and wildfires— have increased in frequency over the past few decades, especially as climate change has increased the likelihood of these events in the present and future (Goss et al., 2020; Michaelis et al., 2022; Williams et al., 2022). Literature suggests that increases in temperature and vapor pressure deficit since 1984 have led to the enhanced dryness of vegetation, which in turn has doubled the forest fire area across the western United States compared to natural climate variability alone (Abatzoglou and Williams, 2016). The situation is even more pronounced in California where beneficial autumnal rains that help reduce fire risk have shifted by almost a month since the 1960s and have concentrated most of the annual precipitation to December-March (Luković et al., 2021). Increasing precipitation variability across the state throughout the 20th century driven by wetter extremes, defined as events higher than the 90th percentile, is further complicating these dynamics (Zamora-Reyes et al., 2022). The confluence of all these variables has resulted in even more complex interactions between the landscape and the availability of the water supply in northern California.

Forested lands in the western United States provide about 65% of the water supply (Brown et al., 2008) and in California, the upper Feather River watershed is an important region for water and hydropower resources (Koczot et al., 2005). Located in the northern Sierra Nevada mountains, this snow-dominated watershed delivers resources to about 27 million Californians and more than 3,000 square kilometers (km²) of farmland through the State Water Project (SWP; California Department of Water Resources, 2025). Drought-inducing fluctuations in precipitation have outsized impacts downstream on water rights allocations for the \$50 billion agricultural industry of the Central Valley and the cities that depend on surface water delivery (Escriva-Bou et al., 2022; Peterson et al., 2023). The California Department of Water Resources (DWR) operates the SWP and traditionally uses snowpack and snow water equivalent (SWE) measurements as a metric to forecast runoff and anticipate water supply for the dry summer months (DWR, 2025). However, after consecutive drought years, dry soils across the watershed act as a sponge, substantially absorbing rain and snowmelt and reducing the amount of water that could have reached reservoirs (Lapides et al., 2022). The upper Feather River watershed has also been affected by large and intense wildfires the last few years, which can alter hydrological processes within the basin (Walters et al., 2024).

The North Complex Fire during the summer of 2020 and the Dixie and Sugar Fires in fall 2021 burned a total of 52% of the upper Feather River watershed (Fig. 1). The North Complex Fire, originally a collection of 21 wildfires that started in the Plumas and Lassen National Forests, was started by lightning from a dry thunderstorm that ignited and burned 1,290 km² from August 17 to December 3, 2020, with 16 fatalities and 2,455 structures destroyed (Safford et al., 2022). About 47% of the vegetation within this wildfire footprint was classified as having high soil burn severity, while 25% had moderate, 24% had low and 5% remained unburned (U.S. Geological Survey et al., 2022). The following year, the Sugar Fire started on July 2, 2021 in the Plumas National Forest while the Dixie Fire ignited a few days later, on July 13, 2021; both fires were

extinguished by October 25, 2021 (Taylor et al., 2022). The soil burn severity distribution within these two wildfires was different from the North Complex Fire; 24% of the area had a high soil burn severity, 29% moderate, 38% low and 12% was unburned (U.S. Geological Survey et al., 2022). The main vegetation type that was burned across all fires was Sierran mixed conifer, consisting of 42% for the North Complex Fire and 51% for the Sugar and Dixie Fire footprints (Fig. S1; CALFIRE, 2015). Other main vegetation types that were burned in the North Complex Fire were White fir (15%), Montane hardwood (12%), and Montane hardwood-conifer (12%; Fig. S1; CALFIRE, 2015). The main vegetation types after Sierran mixed conifer for the Sugar and Dixie Fires were Eastside pine (10%), White fir (9%), and Montane chaparral (6.4%; Fig. S1; CALFIRE, 2015). As of 2024, the Dixie Fire was the second largest (3,750 km²) and the most expensive (\$637 million in suppression costs) fire in California's history (Taylor et al., 2022). Additionally, these fires have negatively impacted water and air quality affecting various metropolitan areas in northern California such as the San Francisco Bay and the Sacramento Valley (Carreras-Sospedra et al., 2024; Dahm et al., 2025).

These major fire events illustrate the lasting effects of past fire-suppression policies of the 1900s, including the lack of fuel management, combined with present-day conditions that promote extreme fire weather and increased fire season length, such as increasing air temperatures, low humidity caused by decreases to autumn precipitation, and dry vegetation (Taylor and Beaty, 2005; Goss et al., 2020; Safford et al., 2021; Taylor et al., 2022; MacDonald et al., 2023). On a shorter time-scale, these major fires were influenced by back-to-back local extreme weather conditions – a wetter than average winter in 2017 and drier than average winters in 2020 and 2021 – where fine fuels (e.g. grasses) had the chance to grow after the wet winters, while the understory vegetation had a prolonged amount of time to dry out during the extended hot and dry springs, summers, and autumns (Taylor and Beaty, 2005; DWR, 2018, 2021, 2022; Swain, 2021). Some studies show no relationship between burn severity and post-wildfire (postfire) runoff (Vieira et al., 2015; Wu et al., 2021), but others show that high burn severity leads to increased runoff and decreased evapotranspiration (ET) postfire (Moazeni and Cerda, 2024), slower plant recovery (Mayor et al., 2007), and decreased infiltration rates (Moody et al., 2016). Moreover, the impact of wildfire extent and severity on the water balance has been understudied (Moazeni and Cerda, 2024).

Wildfires burn and change the vegetation distribution and surface soil properties of an area depending on fire severity and antecedent moisture conditions. For example, vegetation mortality has the potential to change the water balance as ET is reduced (DeBano et al., 1998), though the resulting relationship could be more convoluted depending on the ecosystem (Massmann et al., 2019). A literature review focused on western conifer forests in the United States found that this type of stand-replacing disturbance that decreases ET typically led to increases in streamflow but could also lead to decreases or no changes depending on watershed characteristics postfire such as rapid regrowth and vegetation type (Goeking and Tarboton, 2020). In the western United States, several studies found an increase in runoff following wildfires in semi-arid regions (Kinoshita and Hogue, 2011, 2015; Williams et al., 2022) and a snow-dominated region

within Yosemite National Park in the central part of the state (Boisramé et al., 2019). Wildfires can also lead to soil water repellency (hydrophobicity) to depths of several centimeters that can last several years, dependent on many factors including the burn severity and soil properties (Caltabellota et al., 2022; DeBano, 2000; Doerr et al., 2009; Tinebra et al., 2019). However, results from field studies of hydrophobicity tend to vary postfire; hydrophobicity has been found to often be spatially patchy and non-continuous and can also vary temporally and in intensity (DeBano, 1998). Other impacts from wildfires on the components of the eco-hydrosphere includes loss of surface cover and increases in ash resulting in soil sealing (Larsen et al., 2009), decreased water quality (Rust et al., 2018; Smith et al., 2011; Magliozzi et al., 2024), flash floods, debris flows and hillslope erosion (Blount et al., 2020; Dow et al., 2024; Ebel et al., 2012, 2017; Ebel, 2020; Santi and Morandi, 2013; Saxe et al., 2018).

Recent modeling studies in California have mostly focused on estimating ET through remotely sensed data and quantifying changes to the water balance after wildfires (Ma et al., 2020; Walters et al., 2024). A previous study evaluated the hydrologic impact of the North Complex and Dixie Fires (as well as the Camp Fire in 2018) on the hydrology of the upper Feather River watershed using only remotely sensed data and found increased soil moisture due to decreases in vegetation and ET following the fires (Walters et al., 2024). However, the Normalized Difference Vegetation Index (NDVI) used for the study may not capture vegetation density or health accurately in Evergreen Forests as NVDI tends to not change throughout the year and be somewhat stable (Ukasha et al., 2022). Moreover, it has been suggested that modeling studies focused on the relationship between wildfire and its hydrologic impact should emphasize integration of vegetation regrowth, which may have a stronger influence on streamflow dynamics than climate alone (Bart et al., 2016; Ebel et al., 2023). Other studies have suggested focused understanding of the impact of wildfire on water balance dynamics using quantitative analysis, such as Leaf Area Index (LAI), compared to categorical or qualitative analyses (Goeking and Tarboton, 2020). In evergreen forests, LAI has a seasonal cycle where the lowest values are in the winter and the highest values in the summer (Ukasha et al., 2022).

We applied the Basin Characterization Model (BCM), a physically based water balance model, to quantify changes in evapotranspiration, soil moisture, and streamflow in the upper Feather River watershed due to vegetation changes postfire caused by the North Complex, Sugar, and Dixie Fires. The goals of this study were to quantify: 1) long term trends in climate and hydrology in the upper Feather River watershed, 2) postfire differences in water supply and other water balance variables using dynamic vegetation regrowth over three years in the upper Feather River watershed, and 3) whether the incorporation of postfire vegetation changes in BCM improves streamflow accuracy.

2. STUDY AREA

The upper Feather River watershed in northern California is about 9,300 km² and experiences some of the highest annual precipitation totals in the state with an annual average of 1,117 millimeters (mm; Daly et al., 2008). The precipitation distribution within the watershed has a unique pattern (Fig. 2a); the wettest part of the basin is confined to the western part of the watershed where annual average precipitation can range

from 1,200 mm to more than 2,000 mm (Fig. 2a; Daly et al., 2008). The northern and southeastern parts of the watershed receive on average less than 1,000 mm of precipitation due to the rain shadow effect caused by the Sierra Nevada; the far southeastern corner receives less than 400 mm (Fig. 2a; Fig. S1; Daly et al., 2008). Most of the precipitation accumulates during October-March, but the runoff feeding into Lake Oroville comes from rain and snowmelt in January-June (Zamora-Reyes et al., 2022). Recent studies have suggested that this region continues to experience increased precipitation and streamflow variability complicating intra and interannual management of water resources in the watershed (Swain et al., 2018; Zamora-Reyes et al., 2022). Annual water allocations from Lake Oroville through the SWP provide about 3.95 km³ (3.2 million acre-feet) for drinking water and irrigation (DWR, 2025), which is close to its total capacity of 4.19 km³ (3.4 million acre-feet).

The upper Feather River watershed is crucial to California's water supply; despite only making up 6% of the total area contributing flow to the Sacramento-San Joaquin Delta, it produces 14% of the outflow (CWD and BOR, 2023). The watershed is divided into four subbasins: West Branch Feather River, North Fork Feather River, Middle Fork Feather River, and South Fork Feather River (Fig. 2). The North Fork Feather River comprises the largest area of the watershed and about 60% of the inflow into Lake Oroville while the Middle Fork Feather River contributes 30%, and the smaller West Branch Feather River and South Fork Feather River contribute just 5% (Koczot et al., 2005; Walters et al., 2024). This watershed is highly managed with powerhouses, diversions, and reservoirs that are used for several purposes in the watershed such as hydropower, water supply, recreation and flood control (Koczot et al., 2005). This has resulted in altered discharge measurements at most streamgaging stations in the region making it difficult for modeling studies to use these data for calibrating a natural systems model (Walters et al., 2024).

The upper Feather River watershed has a low population density and remains relatively undisturbed by human development. About 80% of the basin is managed by the U.S. Forest Service and agricultural valleys on the eastern side are privately owned and used for hay production and livestock grazing (George et al., 2007). There is a variety of vegetation types within the watershed, but the most common type is the Sierran mixed conifer (CALFIRE, 2015), which covers about 42.2% of the basin and is an important part of the timber industry in the area (Koczot et al., 2005; Fig. S2; Table S1). Other dominant vegetation types include White fir (9.3%), Eastside pine (8.8%), Sagebrush (5.7%), Chaparral (5.6%), and Grassland (3.5%; CALFIRE, 2015; Fig. S2; Table S1). The northeast and southwest areas of the basin are comprised of the Sierra Nevada and Basin and Range volcanic formations that allow deep percolation due to their high permeability, enhancing contributions toward groundwater flow (Koczot et al., 2005). The central and southeastern parts of the watershed are made up of sedimentary and intrusive igneous formations, which tend to be less permeable than volcanic formations and contribute significantly to surface runoff, subsurface flow, and groundwater recharge (Koczot et al., 2005).

3. DATA AND METHODS

3.1 Basin Characterization Model

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The Basin Characterization Model (BCM; Flint et al., 2021) is a monthly, gridded water balance model that calculates the water balance (precipitation minus evapotranspiration minus sublimation minus runoff minus recharge minus change in storage is equal to zero) using climate inputs: precipitation, minimum temperature, maximum temperature, and potential evapotranspiration (PET). Inputs, water-balance equations, and post-processing schematic are represented in the conceptual BCM diagram (Fig. 3). The upper Feather River BCM is a 30-meter (m) gridded model that was developed by spatially downscaling 4-kilometer (km) monthly Parameter-elevation Regressions on Independent Slopes Model (PRISM; <https://prism.oregonstate.edu>; Daly et al., 2008) climate data (precipitation, minimum and maximum temperature) to 30-m (Flint and Flint, 2012). PET was calculated using air temperature and a modified Priestly-Taylor equation (Flint and Childs, 1991; Priestly and Taylor, 1972), together with topographic shading and cloudiness to calculate hourly PET. More detailed information about BCM calculations and pre-processing can be found in Flint et al. (2021).

Static gridded model inputs include elevation, soil texture and thickness, land use, and geology. A 30-m digital elevation model (DEM) was downloaded from The National Map (USGS, 2022) and clipped to the upper Feather River Watershed. The soil texture data (Batjes et al., 2024), vegetation/land use (California Department of Forest and Fire Protection, 2015), and geology (Horton et al., 2017), were resampled to 30-m and clipped to the upper Feather River watershed.

A historical BCM run was used to calculate monthly water balance output variables from water year (WY) 1896 to 2024. The hydrologic output variables derived by BCM are 1) actual evapotranspiration (AET) which is the water used by plants from soil storage; 2) climatic water deficit (CWD) which is the evaporative demand that exceeds available water (PET minus AET); 3) snow water equivalent (SWE) which is the previous month's accumulated snowpack plus the current snowfall minus the amount of water vapor lost to sublimation and snowmelt; 4) recharge (RCH) or the amount of water that makes it below the zone of ET to become baseflow or groundwater recharge; 5) runoff (RUN) or the excess water that can become streamflow, and 6) soil moisture storage (STR) which is the total amount of water stored in the soil (precipitation plus snowmelt minus AET, RCH and RUN). BCM variable units are millimeters per month, except for temperature, which is degrees Celsius. Baseline AET outputs were calibrated locally to the upper Feather River watershed's dominant vegetation types (Fig. S2; Tables S1 and S2), using gridded AET estimates from Reitz et al. (2017; referred to as ETa) for October 2000 through September 2015. All model codes, control files, input and output layers, and user's manual can be found in the "Upper Feather River Watershed 30-meter Basin Characterization Model – Monthly Historical Climate and Hydrology" data release and model archive (Stern et al., 2026). To further validate and qualitatively compare BCM estimates of ET, annual MODerate Resolution Imaging Spectroradiometer (MODIS) ET maps were downloaded from <https://earthdata.nasa.gov/sensors/modis> (Running et al., 2021). Monthly BCM maps of AET were summed over each calendar year from 2020-2023 for comparison against the MODIS evapotranspiration maps, but all other annual comparisons were done by WY (October through September).

Monthly streamflow data from 21 USGS (USGS, 2024) and two California Data Exchange Center (CDEC, 2023; <https://cdec.water.ca.gov>) stations were downloaded to calibrate simulated BCM streamflow from October 1896 to September 2024 (see Table 1 for station details). Since many USGS streamgages in the watershed are affected by human alteration, only data from stations where diversion data was included and, in regions with dams or reservoirs, only pre-dam data was used (Table 1). CDEC stations are full-natural flow estimates, which are a modeled representation of unimpaired streamflow calculated by DWR (Huang and Kadir, 2017). The period of record used was not consistent across all gages (Table 1) and all available data was used for calibration with none for validation, as previous studies have suggested that it is not a necessary step in hydrologic modeling (Table S3; Shen et al., 2022). Simulated flows at each streamgage were post-processed using recharge and runoff outputs summarized by subbasin. Streamflow parameters were adjusted manually until the fit with measured data was optimized, and calibration goodness-of-fit statistics were maximized (Table S3). The calibration statistics used included the coefficient of determination (R^2), the Nash-Sutcliffe coefficient of efficiency (NSE) and percent bias (PBIAS); these are standard metrics used to quantitatively compare observed and simulated flow (Stern et al., 2016). To route upstream flow, individual subbasin flows were added to downstream subbasins. For more details about these methods, see the BCM Techniques and Methods report (Flint et al., 2021).

3.2 Wildfire and Vegetation Index input

Thematic Burn Severity Classification rasters derived from continuous Normalized Burn Ratio images were downloaded from Monitoring Trends in Burn Severity (MTBS, <https://mtbs.gov>; Eidenshink et al., 2007) for each wildfire at 30-m resolution. The burn severity maps were calculated using Landsat remote sensing data and categorized into six burn severity classes: unburned to low, low, moderate, high, increased greenness and a non-processing mask (Eidenshink et al., 2007). These rasters were then used to develop maps of vegetation change postfire as inputs to BCM. Vegetation Index maps (also called the LAI parameter in BCM) used in BCM represent an index of vegetation health, and track vegetation growth and die back through time as a function of climate variability (Flint et al., 2021). The Vegetation Index maps are output from BCM at a monthly time step and can be replaced to simulate vegetation disturbance (Flint et al., 2021; Flint et al., 2024). To create Vegetation Index maps that accounted for changes in vegetation due to wildfires, each MTBS burn severity grid cell was replaced using the following conversions: unburned to low severity = 0.75, low severity = 0.50, moderate severity = 0.25, and high severity = 0.00 (Flint et al., 2021). These new Vegetation Index maps were input into BCM the month after the wildfire occurred to reduce vegetation density for the month postfire. After the burn severity-based Vegetation Index map was input, vegetation was allowed to grow and die back based on changes in climate, but no other BCM parameters were changed. BCM uses a growth rate parameter that determines how quickly vegetation is allowed to grow back, based on the amount of precipitation received given a particular month relative to a long-term average (instead of average annual precipitation in normal mode; Flint et al., 2021). Areas where the Vegetation Index is less than 1.00, the BCM reverts to a disturbance mode where the vegetation parameters

allow recovery of vegetation based on calibrated vegetation parameters to represent postfire regrowth (Flint et al., 2021). Once the Vegetation Index returns to 1.00, normal vegetation calculations continue based on original vegetation type parameters. A baseline simulation with same background climate data was run for the same period except that it did not ingest the modified Vegetation Index maps; this simulation was used for a baseline (no fire) comparison.

Baseline and postfire AET, soil moisture, and streamflow were analyzed and compared for two stations and their contributing subbasins upstream as well as the full natural flow estimate at Feather River at Oroville (Fig. 4, green dots; Table 1). To quantify calibration changes due to model parameterization, baseline and postfire streamflow statistics were calculated separately for the baseline (2020-2024) and postfire (2020-2024) against measured and reference data (Table 3). Percent difference of postfire-to-baseline at each streamflow station as well as runoff coefficients (annual streamflow divided by annual precipitation) were calculated to further understand the streamflow response to wildfires (Table 4). Differences in baseline and postfire AET and soil moisture for three different groups were calculated using burned area thresholds within a subbasin (Fig. 4): subbasins with total burned area <25%, subbasins with total burned area of >25% in the North Complex Fire in 2020 (Table S4), and subbasins with total burned area of >25% in the Dixie and Sugar Fires in 2021 (Table S5).

4. RESULTS

4.1 Historical long-term climate and hydrology trends

BCM was used to generate average annual (WY 1991-2020) SWE (Fig. 2b), runoff (Fig. 5a), recharge (Fig. 5b), and runoff plus recharge (water supply; Fig. 5c). As expected, the average annual SWE in the upper Feather River watershed (Fig. 2b) follows a similar pattern to precipitation (Fig. 2a) where the highest amounts are in the western part of the basin and where elevations are the highest (Fig. S1). These areas accumulated more than 1,000 mm in SWE, but the whole watershed (excluding low-lying elevations) received more than 50 mm of SWE on average, illustrating the importance of snow to the water supply (Fig. 2b). The highest average annual recharge and runoff patterns are similar to precipitation and SWE, except that most of the recharge and runoff is concentrated in the southwestern part of the upper Feather River watershed (Fig. 5). One notable difference between these two variables in this part of the basin is that average runoff tends to be higher in magnitude and covers a larger geographical area; the average annual runoff ranges from 600 mm to 1,700 mm (Fig. 5a) while average annual recharge is mostly around 600 mm with some small areas reaching 1,000 mm (Fig. 5b). These spatial patterns coincide with underlying bedrock permeability where areas with volcanic formations and high permeability have the highest recharge, whereas those regions with intrusive igneous formations have lower permeability and higher runoff (Koczot et al., 2005). Regardless of the breakdown between these two individual variables, the western area of the watershed has the highest water supply potential (Fig. 5c).

A three-year running average of climatic water deficit (CWD or potential evapotranspiration minus actual evapotranspiration), a measure of water stress in the upper Feather River watershed is shown in Fig. 6. CWD has fluctuated since WY 1896 through dry and wet periods, however the lowest and highest values were generated during the last 40 years; CWD was less than 350 mm per year in 1984 and has been on a steady rise since 1999 attaining the highest value in 2023 (Fig. 6). This reflects the enhanced ability of the atmosphere to dry out vegetation as air temperatures continue to increase. CWD appears to have become more variable over time, particularly after the mid-1970s (Fig. 6). Additionally, long-term average annual analysis of precipitation comparing values from 1981-2020 show that the southwestern part of the upper Feather River watershed has experienced a decrease of up to 11 mm per year; in fact, most of the basin saw a decrease except for a minimal increase to the southeastern side of the basin (Fig. S3a). The same historical analysis comparing SWE values shows a more complex spatial pattern as higher elevations have experienced a 5 to 10 mm per year decrease whereas low elevations showed minimal change (Fig. S3b).

4.2 Precipitation and snow water equivalent conditions before, during, and after wildfires

Spatial and temporal analyses of precipitation (Fig. 7) and SWE (Fig. S4) for WY 2020 to 2024 revealed important context with respect to long-term conditions described in section 4.1. During the winter of 2020, precipitation and SWE within the upper Feather River watershed were about 41.5% and 67.2% below the historical average from 1991-2020, respectively, leading up to the North Complex Fire during the summer of 2020 (Fig. 7a and S4a). These deficits were more pronounced the following year (2021) as precipitation dipped to 55.5% of average (Fig. 7b) and SWE was 86.6% of average (Fig. S4b); the Sugar and Dixie Fires occurred a few months later in summer and fall of 2021. WY 2022 provided little replenishment to the area; precipitation was about 82.2% of average (Fig. 7c) and SWE remained at very low levels with 13.9% of average (Fig. S4c). After three years of consecutive dry conditions, the region saw a major surplus in precipitation (Fig. 7d) and SWE (Fig. S4d) in 2023; precipitation was 27.2% above average while SWE was 181.3% above average. The last year included in this analysis (2024) had conditions that were closer to long-term average: precipitation was slightly below average (87.2%; Fig. 7e) while SWE was slightly above average (106.9%; Fig. S4e).

4.3 Postfire changes in AET, soil moisture, water supply, and streamflow

Modeled output from the two calibration stations, Feather River at Merrimac, which captures upstream changes caused by the North Complex Fire in 2020, and North Fork Feather River below Rock Creek, which captures upstream changes to the North Fork Feather River caused by the Sugar and Dixie Fires in 2021, showed decreases in actual evapotranspiration (AET; Fig. 8a) and increases in soil moisture (Fig. 8b) postfire.

The Feather River at Merrimac subbasin experienced changes in AET right after the North Complex Fire from the first through the fourth year postfire; the highest changes in AET occurred the second year after the fire (Fig. 8a, dotted lines). For the North Fork Feather River below Rock Creek subbasin, AET changes

334 were seen immediately after the Sugar and Dixie Fires in 2021, but the most change occurred the year
 335 after, during the summer of 2022 (Fig. 8a, solid lines). The average seasonal change in AET from WY 2021-
 336 2024 in the Feather River at Merrimac subbasin decreased -9% (-3.2 mm) during fall (October-December),
 337 to -10% (-6.6 mm) during winter (January-March), -11% (-26.5 mm) in spring (April-June) and -9% (-17.8
 338 mm) in summer (July-September). North Fork Feather River below Rock Creek resulted in higher
 339 decreases: -9% (-3.9 mm) during fall, -13% (-9.1 mm) in winter, -16% (-42.8 mm) in spring and -16% (-38.5
 340 mm) in summer. Maximum decreases in AET for both subbasins occurred during the summer of 2022 with
 341 -13% and -26% for the Feather River at Merrimac and North Fork Feather River below Rock Creek
 342 subbasins, respectively.

343 Increases in soil moisture changes were seen throughout the whole year, with the largest increases
 344 occurring between August-December 2022 (Fig. 8b). There was minimal change in soil moisture
 345 immediately following the fires, but the year after showed major increases for both subbasins, particularly
 346 for North Fork Feather River below Rock Creek (Fig. 8b, solid lines). At Feather River at Merrimac, soil
 347 moisture differences were maximized two years postfire following the North Complex Fire (Fig. 8b, dotted
 348 lines). The average seasonal change in soil moisture from WY 2021-2024 in the Feather River at Merrimac
 349 subbasin increased from 7% (51.4 mm) during fall, to 2% (18.3 mm) during winter, 3% (29.2 mm) in spring,
 350 and 9% (76.6 mm) in summer. For North Fork Feather River below Rock Creek, values were higher: 12%
 351 (172.9 mm) during fall, 10% (161.5 mm) for winter, 5% (97.9 mm) in spring and 14% (203.9 mm) in summer.
 352 The maximum changes in seasonal soil moisture for both subbasins occurred during fall 2022 (16% or 98.5
 353 mm and 31% or 373.4 mm). However, there were also large increases in winter 2023 for North Fork Feather
 354 River below Rock Creek (27% or 363 mm).

355 Changes in postfire AET and soil moisture led to changes in runoff plus recharge (water supply; Fig. 9).
 356 During WY 2021, water supply within the regions that burned in the upper Feather River watershed had a
 357 25% increase compared to the baseline simulation or an increase of about 2% of average (Fig. 9a and
 358 Table 2). These increases continued through WY 2022 (Fig. 9b) and peaked during WY 2023 (Fig. 9c)
 359 when the water supply postfire increased 34% of average (Table 2). In WY 2024, a close to average year
 360 with respect to precipitation and SWE, water supply postfire continued to be above baseline (23% of
 361 average; Table 2). Overall, the water supply after WY 2021 through WY 2024 showed an increased
 362 potential for streamflow generation within the North Complex and Sugar and Dixie Fire footprints.

363 Postfire streamflow resulted in higher values than baseline at Feather River at Merrimac, North Fork
 364 Feather River below Rock Creek, as well as Feather River at Oroville (Fig. 10). Immediately following the
 365 fires, there were only minimal differences between postfire and baseline after the North Complex Fire in
 366 Feather River at Merrimac (Fig. 10a) and the Sugar and Dixie Fires in North Fork Feather River below Rock
 367 Creek (Fig. 10b). The largest changes in streamflow occurred in December 2022 for Feather River at
 368 Merrimac (161% increase) and Feather River at Oroville (159% increase; Fig. 10c) whereas in North Fork
 369 Feather River below Rock Creek the largest change occurred in March 2023 (101%). The average seasonal

percent change in streamflow from WY 2021-2024 in the Feather River at Merrimac subbasin ranged from 39% during fall, to 19% during winter, 7% in spring and 1% in summer. For North Fork Feather River below Rock Creek, increases were 11% during fall, 35% for winter, 18% in spring and 13% in summer. Seasonal percent changes in streamflow for Feather River at Oroville were higher than baseline conditions: 39% during fall, 19% for winter, 10% in spring and 5% in summer. The largest seasonal changes occurred during fall 2022 at Feather River near Merrimac (64%) and Feather River at Oroville (79%) while in North Fork Feather River below Rock Creek this change occurred during winter 2023, two years postfire (73%; Fig. S5).

Overall, the incorporation of wildfire effects on vegetation into BCM resulted in improved model streamflow statistics (Table 3). All three stations showed increases to both monthly R^2 (between 0.04 and 0.07) and NSE (between 0.05 and 0.18) and decreases in percent bias (between -2.7% and -15.2%). The percent difference of postfire-to-baseline annual streamflow was highest two years postfire for both subbasins: the Feather River at Merrimac annual flows increased by 20% in WY 2022, while North Fork Feather River below Rock Creek saw an increase of 37.1% in WY 2023 (Table 4). Feather River at Oroville, which reflects changes accumulated over the whole watershed, saw increases in streamflow compared to baseline of 3.5% in WY 2021, 24.4% in WY 2022, 20.8% in WY 2023 and 16.2% in WY 2024 (Table 4). The hydrologic response in all three streamflow stations postfire is captured in the runoff coefficient; this metric shows higher values from WY 2021 through 2024 postfire compared to baseline, but in particular during the anomalous wet WY 2023 (Table 4).

4.4 Postfire changes in AET and soil moisture by burn threshold

This study assessed the impact of wildfires on the water balance using a 25% threshold, where subbasins that were less than 25% burned were considered “unburned” and those that were more than 25% burned were considered “burned”, and separated by burn year (2020 or 2021; Tables S4 and S5). The separation of these subbasins into unburned or burned resulted in similar patterns to the analysis focused on total upstream areas from Feather River at Merrimac and North Fork Feather River below Rock Creek gages where decreases to AET (Fig. 11a) and increases to soil moisture (Fig. 11b) were captured postfire.

The difference from baseline for AET after the North Complex Fire (Fig. 11a, dashed lines) and after the Sugar and Dixie Fires (Fig. 11a, solid lines) showed the highest change after the first and second year postfire (about -30% or -7 mm for areas burned in 2020 and almost -32% or -15 mm for areas burned in 2021) while the unburned areas (Fig. S6a) showed minimal differences compared to baseline. The average seasonal change in AET from WY 2021-2024 for the subbasins burned in 2020 saw values of 21% (-2.2 mm) during fall, -28% (-4.5 mm) for winter, 26% (-16.4 mm) in spring, and 22% (-12.7 mm) in summer. Subbasins burned in 2021 average seasonal change for fall was -11% (-2.1 mm), -15% (-4.2 mm) for winter, 18% (-21.8 mm) in spring, and -20% (-19.5 mm) in summer. On an annual basis, the baseline-to-postfire change in AET for the combined annual averages of subbasins burned in 2020 and 2021 were -16% (-5.4

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3 405 mm) in WY 2021, -26% (-14.1 mm) in WY 2022, -19% (-11.8 mm) in WY 2023, and -19% (-10.7 mm) in WY
4 406 2024.

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7 407 There were minor changes in soil moisture immediately postfire in areas burned in 2020 (Fig. 11b, dashed
8 408 lines) and 2021 (Fig. 11b, solid lines). Soil moisture changes peaked during fall of 2022 when both burned
9 409 areas had major increases above 30% or more than 35 mm for areas burned in 2020 and close to 70 mm
10 410 for areas burned in 2021 (Fig. 11b), while unburned areas remained relatively unchanged (Fig. S6b). The
11 411 average seasonal change in soil moisture from WY 2021-2024 in the unburned areas ranged from 3% (15.6
12 412 mm) during fall, 2% (14.5 mm) in winter, 2% (13.9 mm) in spring, and 4% (21.8 mm) in summer. Areas
13 413 burned in 2020 saw higher values: 11% (47.4 mm) during fall, 2% (9.3 mm) for winter, 5% (23.8 mm) in
14 414 spring, and 19% (69.7 mm) in summer, while the highest average seasonal values were found in areas
15 415 burned in 2021: 13% (85.6 mm) during fall, 10% (76 mm) for winter, 5% (47.3 mm) in spring, and 16%
16 416 (102.3 mm) in summer. On an annual basis, the baseline-to-postfire change in soil moisture for the
17 417 combined annual averages of subbasins burned in 2020 and 2021 was 5% (20.4 mm) in WY 2021, 10%
18 418 (58.4 mm) in WY 2022, 15% (85.6 mm) in WY 2023, and 10% (66.3 mm) in WY 2024.

24 419 The wildfire footprints within the upper Feather River watershed differed between 2020 and 2021; the North
25 420 Complex Fire burned about 1,162 km² while the Sugar and Dixie Fires burned 3,014 km², or more than
26 421 double the North Complex Fire. Another difference between these fires is that the North Complex Fire was
27 422 a higher severity than the Sugar and Dixie Fires as the majority of the subbasins had a moderate to high
28 423 severity burn (Tables S4 and S5). The most common vegetation types that burned during the North
29 424 Complex and Sugar/Dixie Fires were the Sierran mixed conifer (42% and 51%, respectively) and White fir
30 425 (15% and 9%, respectively), which are similar to the overall vegetation composition of the upper Feather
31 426 River watershed (Table S1). One difference between the vegetation types burned between the two fires is
32 427 that the North Complex Fire had a higher percentage burned area of Montane hardwood and Montane
33 428 hardwood-conifer (almost 24% compared to less than 4% for Sugar/Dixie Fires; Table S1); this type of
34 429 vegetation has higher water use than Sierran mixed conifer and White fir (Flint et al., 2024).

41 430 **4.5 Comparison of evapotranspiration between MODIS and BCM**

43 431 MODIS ET (500-m resolution) data was used as qualitative validation and comparison to BCM AET (30-m
44 432 resolution) output and showed that, in general, BCM AET tended to be higher than MODIS for WYs 2020-
45 433 2023 (Fig. 12; Fig. S7). This difference is more acute in 2022 where the average MODIS ET was 477.9 mm
46 434 and 511.9 mm for BCM (Fig. 12c, g; Fig. S7b). The lowest ET values for MODIS and BCM occurred in 2021
47 435 (445.3 mm and 412.2 mm; Fig. S7b) while the highest ET values occurred in 2023 (538.2 mm and 563.3
48 436 mm; Fig. S7b). The footprints of the North Complex and Dixie Fires were visible in both ET results, but
49 437 MODIS ET showed a faster recovery of AET by WY 2023 compared to BCM AET estimates.

54 438 **5. DISCUSSION**

This multi-year process-based modeling study reveals that the water balance within the upper Feather River watershed has undergone major changes since summer 2020 due to the impact of some of the largest wildfires in California's history, which have a combined burn area of 4,800 km² or 480,000 hectares equaling more than half of the basin. The study used two streamflow stations for comparison: Feather River at Merrimac, which represents changes to the Middle Fork of the Feather River watershed, and the North Fork Feather River below Rock Creek, which represents changes to the largest contributor of streamflow to Lake Oroville, the North Fork of the Feather River. The Feather River at Merrimac was predominantly affected by the North Complex Fire, and the North Fork Feather River below Rock Creek gage was mostly affected by the Dixie and Sugar Fires. Both subbasins showed increases in AET and soil moisture compared to baseline conditions directly after the fires occurred, but most of the change occurred in the spring and summer for AET and fall, winter and summer for soil moisture 1 to 4 years postfire (Fig. 8). A similar pattern in streamflow, where differences were maximized during the fall, continued throughout the winter and spring and minimized by summer, was also shown (Fig. 10 and Fig. S5). This is consistent with other studies in the Sierra Nevada that have shown postfire increases in spring snowmelt streamflow with lesser changes in summer baseflow (Boisramé et al., 2019). Although these results indicate a decrease in ET that results in higher streamflow and baseflow similar to a recent study in the western United States (Furniss et al., 2025; Williams et al., 2022), other studies have found that wildfire can reduce late-season streamflow via reduced snowpack retention (Kampf et al., 2022; Koshkin et al., 2022) or increased ET from rapid post-disturbance growth or increased evaporation (Goeking and Tarboton, 2020). While the upper Feather River watershed may have begun to slowly recover from these wildfires (Fig. 12), it might be years to decades until it may return to pre-fire conditions (Blount et al., 2020; Niemeyer et al., 2020), especially in high burn severity areas. Moreover, this relatively unmanaged watershed may not return to pre-fire conditions when vegetation was burned with high severity, and therefore may continue to experience decreased AET and increased soil moisture and streamflow into the future (Kinoshita and Hogue, 2011; Ma et al., 2020; Niemeyer et al., 2020; Wells et al., 2024; Williams et al., 2022).

The period between December 2022 through March 2023 was one of the wettest on record in California (Fig. 7d) and occurred after an intense dry period from 2020 through 2022 (Fig. 7a-c). This region received an average precipitation of 1,420 mm during WY 2023 (127% of average) while WY 2020, 2021, 2022, and 2024 received much less with 653 mm (58% of average), 497 mm (48% of average), 919 mm (82% of average), and 974 mm (87% of average), respectively. Snow water equivalent, an important measure of how much water is stored as snowpack for use during the dry season, followed a similar pattern during these five years (Fig. S4). These extreme climatic conditions are reflected through the high CWD experienced in the upper Feather River watershed leading up to the North Complex, Sugar, and Dixie fires (Fig. 6). The increased precipitation conditions in WY 2023 throughout the basin led to the saturation of soils in the area which were also experiencing lower AET due to postfire changes in vegetation cover. During this same period, postfire AET was lower than the baseline, suggesting that the vegetation had not returned to pre-fire conditions (Niemeyer et al., 2020).

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3 476 The combined effects of decreased vegetation after North Complex, Sugar, and Dixie Fires plus record
4 477 precipitation during WY 2023 season led to increased water supply potential in areas located within the
5 478 wildfire footprint (Fig. 9c) and further analysis suggested higher streamflow magnitudes postfire at all three
6 479 comparison locations, particularly in the North Fork Feather River below Rock Creek and the outlet station,
7 480 Feather River at Oroville (Fig. 10b and c). The inclusion of vegetation changes due to wildfires within the
8 481 watershed resulted in a water supply surplus of 1.2 km³ in WY 2023 alone (1 million acre-feet; Table 2),
9 482 which equates to almost a third of Lake Oroville's capacity, led to an increase of 21% more streamflow in
10 483 the upper Feather River basin that year (Table 4). Such "savings" continued into WY 2024 where the surplus
11 484 in the water supply was 8.5×10⁸ m³ or 0.8 km³ (690,000 acre-feet; Table 2) and about 16% more streamflow
12 485 into Lake Oroville (Feather River at Oroville; Table 4). It is hard to separate the individual contribution
13 486 between the wildfires and the extraordinary precipitation in the WY 2023 season as it has become more
14 487 difficult to assess and model an "average" winter in California due to the high hydroclimatic variability in the
15 488 region (Zamora-Reyes et al., 2022). However, in the western United States this relationship becomes non-
16 489 linear (Williams et al., 2022) and could be further studied in this watershed. Previous efforts to quantify water
17 490 balance dynamics postfire in the upper Feather River watershed relied on remote sensing only and did not
18 491 analyze streamflow data due to the highly managed system (Walters et al., 2024), missing an important
19 492 element toward a more comprehensive understanding of hydrological processes in the region. As
20 493 precipitation variability throughout the state continues to increase into the 21st century, it is important for
21 494 future studies to focus on improving the understanding of the relationship between extreme precipitation
22 495 events (Swain et al., 2018; Touma et al., 2022; Zamora-Reyes et al., 2021), and the long-term effects of
23 496 the North Complex and Dixie Fires on runoff, soil moisture, and actual evapotranspiration in the upper
24 497 Feather River watershed.

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26 498 Previous studies have indicated possible wildfire burn thresholds at which changes in postfire vegetation
27 499 coverage and intensity are significant enough to produce changes in the water balance (Saxe et al., 2018).
28 500 Although there is some consensus in the literature of a 20% wildfire burn threshold within a watershed to
29 501 see major hydrologic change (Hallema et al., 2018; Saxe et al., 2018; Williams et al., 2022; Wine et al.,
30 502 2018), this study used a threshold of 25% or more burned area since there was very little change between
31 503 baseline and postfire model runs between unburned areas (25% or less burn area) compared to those that
32 504 burned (greater than 25% burn area) during 2020 and 2021 (Fig. S6 vs Fig. 11). The results reinforced the
33 505 idea that there was not enough of a disturbance to see significant changes within those basins below the
34 506 threshold. For the basins that exceeded this threshold, changes in AET and soil moisture were highest in
35 507 the second year postfire (Fig. 11) and may likely continue to impact water resources into the future due to
36 508 the size and severity of the North Complex and Dixie Fires that burned over half of the watershed. Across
37 509 the upper Feather River watershed, 29% was moderately burned and 31% was severely burned. Studies
38 510 in adjacent watersheds, the Yuba River and American River, which have similar vegetation to the upper
39 511 Feather River watershed, examined wildfires between 1985-2018 and found that medium-intensity wildfires
40 512 (defined in Roche et al., 2020 as those with a loss in basal area between 25-75%) in that period did not

have a significant effect on streamflow even though they caused large ET changes due to vegetation cover reduction, which could also be explained by the size of the wildfires relative to the size of the watershed (Roche et al., 2020). The largest wildfire included in that study was 8% of the watershed (the American River watershed area is 4825 km² and the King Fire in 2014 burned: 373 km² of the watershed; Roche et al., 2020), which is much smaller than North Complex or Dixie Fires and the 25% threshold. Thus, the upper Feather River watershed and the wildfires of 2020 and 2021 are a unique case study postfire hydrology due combined effects of the watershed size and the percent burned area, as no other studies of comparable watershed or wildfire size were found at the time of writing this manuscript.

This study highlights the flexibility of BCM for investigating hydrological changes due to disturbances such as wildfires that may be difficult to directly measure. One of the major advantages of this model is its ability to incorporate bedrock conductivity, an often-ignored variable in rainfall-runoff models, that can lead to better recharge and runoff estimates (Flint et al., 2021; Flint et al., 2024). BCM integrates vegetation parameters, such as growth rates and vegetation health, that have been calibrated for 72 different vegetation types in California; it should be noted that the baseline BCM version does not include any of the wildfire effects. A recent literature review suggested that incorporating vegetation growth and health are important to include in postfire modeling studies to better understand associated hydrologic changes (Ebel et al., 2023). Others have argued that physically based models will become increasingly important to understand hydrological processes postfire (Partington et al., 2022) and that future studies on postfire processes should include the ability to quantify at least annual precipitation, annual maximum snow water equivalent, and magnitude of hydrologic change (Goeking and Tarboton, 2020).

Water balance models are powerful investigative tools that can be used to simulate historical and future hydrological changes due to climate and land use yet have inherent limitations that should be considered when interpreting results from this study. BCM simulates snowpack processes using the SNOW-17 model, which does not have the capability to model snow interception or vegetation impacts on snow attenuation (Anderson, 2006). Although snowpack was calibrated to snow pillow and snow course data across California (CDEC, 2025), this model assumption may lead to higher uncertainty in snow accumulation and melt processes and the seasonality of snowpack. BCM is an unimpaired watershed model, which means human-induced changes like diversions, urbanization, logging, and other disturbances are not explicitly modeled. BCM does not include streamflow routing directly, as streamflow is calculated using a post-processing spreadsheet that accumulates recharge and runoff for all grid cells upstream of a gage location (Flint et al., 2021). This may be a limitation if there are significant surface water-groundwater interactions above and below the gage location or if ungaged diversion data are not available. In addition, lateral flows are not simulated, which may or not be a large source of uncertainty depending on local soil, topographic, and geologic conditions. BCM also does not simulate loss of surface cover, soil sealing, ash layers, or hydrophobicity (where soil particles are sometimes affected by wildfire to repel water), which effect runoff generation. Many studies have shown that soil conditions may be hydrophobic before wildfires, that

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3 549 hydrophobicity is non-continuous across a landscape, or that soils may become hydrophobic postfire.
4 550 Therefore, the runoff processes may be enhanced or reduced, depending on the local soil properties and
5 551 soil burn severity, adding to the uncertainty of the projected changes in streamflow. Without additional soil
6 552 data, it is not possible to accurately characterize the soil hydrophobicity or soil sealing across the
7 553 watershed, and in this study, the soil conditions were assumed to be the same pre and postfire.
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11 554 Another limitation to this study is the lack of measured ET and soil moisture data to validate BCM outputs.
12 555 However, the comparison between MODIS ET and BCM AET showed that BCM postfire simulations
13 556 captured the changes in ET due to the 2020 and 2021 wildfires (Fig. 12). BCM annual ET was higher
14 557 compared to MODIS (Fig. S7), yet previous studies have suggested that MODIS generally underestimates
15 558 ET (Mu et al., 2011). Previous wildfire footprints including the Walker Fire in 2019 and the Camp Fire in
16 559 2018 were visible in the MODIS ET and likely affected watershed processes but were not covered in this
17 560 study since they occurred before 2020. Future work could incorporate all major recent fires including the
18 561 Camp and Walker Fires, and several smaller fires that occurred over the last ten years in the watershed.
19 562 Very few studies have evaluated soil moisture model performance against instrumental data or used it for
20 563 initial conditions (Ebel et al., 2023), and more soil moisture data could help to fill this information gap in this
21 564 watershed. Incorporating additional soil moisture data may help calibrate the model to improve the
22 565 understanding of drying and wetting cycles in a snow-dominated location and the current understanding of
23 566 high runoff production areas throughout the watershed. BCM's vegetation growth parameters could also be
24 567 calibrated against soil moisture instrumental data to incorporate subsequent adjustments to the vegetation
25 568 caused by climatic variation that may not be adequately represented by the fire effects. Further work could
26 569 also address vegetation type conversion, which is not accounted for in this study. Insight into these
27 570 processes is critically important for water resources management and streamflow forecasting.

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36 571 **6. CONCLUSIONS**

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38 572 The cascading impacts of wet and dry extreme climatic events in California, such as droughts, floods, and
39 573 wildfires, are directly affecting the state's water supply. This study was the first to quantify the changes to
40 574 the monthly, seasonal, and annual actual evapotranspiration (AET), soil moisture, and streamflow changes
41 575 in the upper Feather River watershed following the North Complex Fire in 2020 and the Dixie and Sugar
42 576 Fires of 2021 using a process-based model. This study used the Basin Characterization Model, a gridded
43 577 water balance model, which allowed for the dynamic integration of vegetation change resulting from the
44 578 wildfires. The combined effects of decreased AET after the North Complex, Sugar, and Dixie Fires with
45 579 record precipitation during the water year (WY) 2023 season led to enhanced soil moisture levels and higher
46 580 streamflow postfire at all three streamflow stations used in the study; the upper Feather River basin saw
47 581 32% higher streamflow during WY 2023 and 16% in WY 2024. Although the majority of this change occurred
48 582 during the wet season, its effects were evident during the dry season as increases to summer baseflow of
49 583 12% in WY 2023 and 2% in WY 2024.

These results are important for local, state, and federal agencies that monitor and manage the water supply in the state to consider since relatively unmanaged basins affected by intense wildfire, such as the upper Feather River watershed, are likely to continue to experience above normal streamflow due to vegetation changes for years to decades (Blount et al., 2020; Niemeyer et al., 2020). Studies have shown that forest thinning and controlled burns are promising methods for enhancing water supply resilience (Loverin et al., 2024; Robles et al., 2014; Boisramé et al., 2017; Sun et al., 2015). There may be a buffering effect to this level of disturbance on the watershed, and other studies have shown there are potential benefits of managing forests to maintain water supply resilience (Niemeyer et al., 2020; Boisramé et al., 2017; Sun et al., 2015). However, a generic approach to forest management for water supply can cause unintended consequences like increasing forest water use and drought vulnerability (Tague et al., 2019).

Although results presented in this paper indicate that wildfires may benefit the water supply, predictions of increased precipitation variability throughout California can leave the state in a more vulnerable situation that may require flexible adaptation strategies to reduce flood risk (Swain et al., 2018; Touma et al., 2022; Zamora-Reyes et al., 2022). Future work could include additional validation of BCM soil moisture outputs in the upper Feather River watershed against in-situ soil moisture and evapotranspiration data to improve the understanding of runoff generation in the region. Studies have suggested continued management of mixed-conifer forested lands in upstream watersheds that supply water to sustain these water benefits and are likely to become increasingly important through climate change, disturbances like wildfire and landslides, and other land use changes (Boisramé et al., 2019). The upper Feather River watershed is one important piece of the complex water system in California that could benefit from a coordinated effort between several agencies across the state to manage resources for wildlife, municipal and residential water supply, and agricultural use into the future.

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CONFLICT OF INTEREST

There is no conflict of interest for any of the authors. Any use of trade, firm, or product names is for descriptive purposes only and does not imply endorsement by the U.S. Government.

DATA AVAILABILITY STATEMENT

All model codes, control files, input and output layers, and user's manual are available on "Upper Feather River Watershed 30-meter Basin Characterization Model – Monthly Historical Climate and Hydrology" data release and model archive (Stern et al., 2026). Streamflow data for U.S. Geological Survey stations can be found in the USGS Water Data for the Nation website (<https://waterdata.usgs.gov>; USGS, 2024) and for California Department of Water Resources stations in the California Data Exchange Center (<https://cdec.water.ca.gov>; CDEC, 2025). MODerate Resolution Imaging Spectroradiometer (MODIS) evapotranspiration data are available in the MODIS website (<https://earthdata.nasa.gov/sensors/modis>; Running et al., 2021). Burn Severity Classification rasters can be downloaded from the Monitoring Trends in Burn Severity website (<https://mtbs.gov>; Eidenshink et al., 2007).

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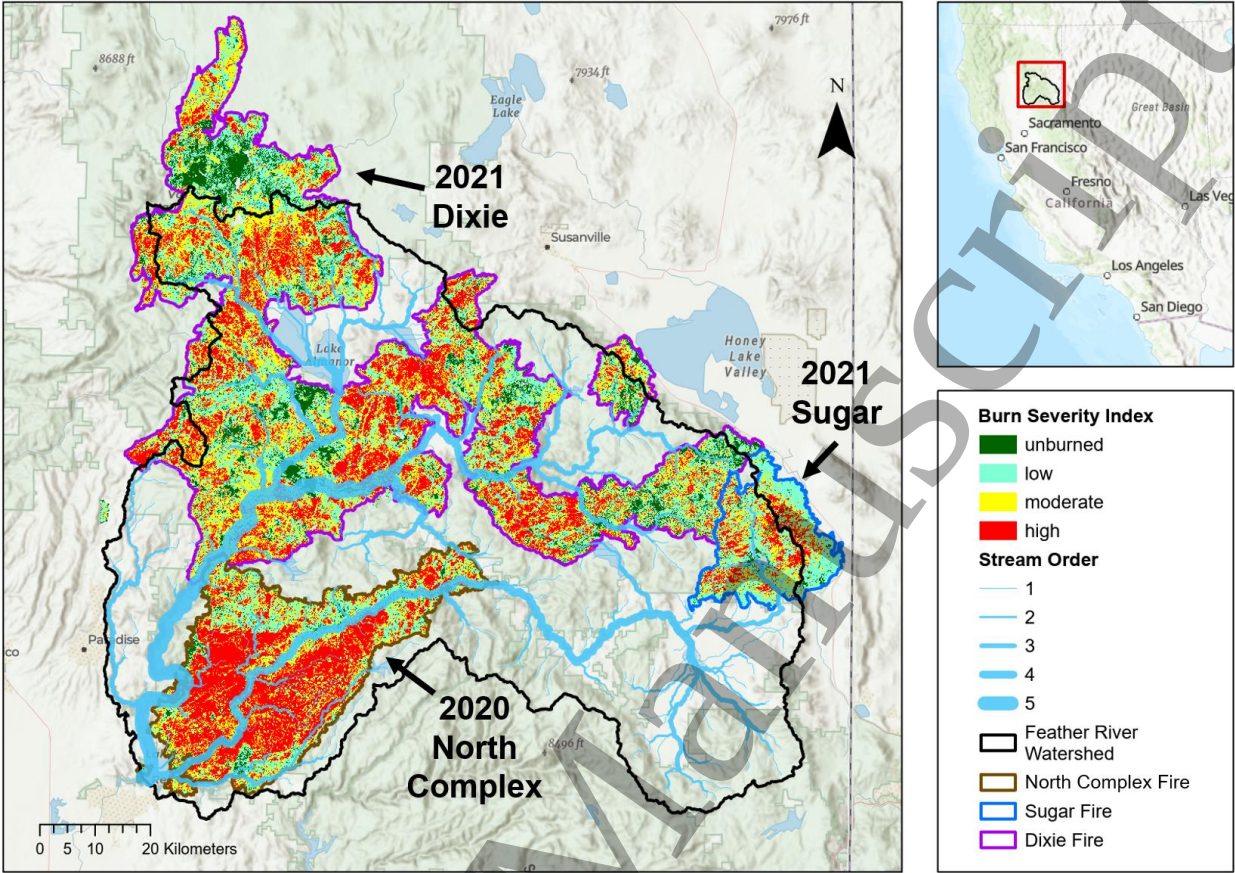
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Figure 1: Study area map of the upper Feather River watershed with the Burn Severity Index after the North Complex (brown outline) Fire in 2020 and the Sugar (blue outline) and Dixie (purple outline) Fires in 2021; Burn Severity Index raster data were downloaded from the Monitoring Trends in Burn Severity website (<https://mtbs.gov>). Also shown are Stream Order (aqua lines) which were produced using the USGS 30-meter digital elevation model.

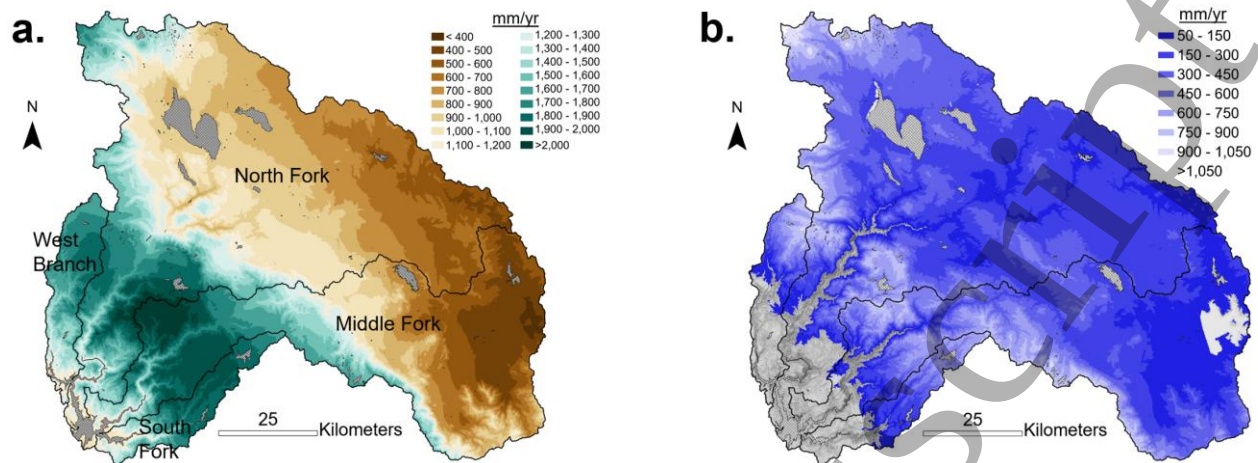


Figure 2: Average annual precipitation (a) and snow water equivalent (b) in the upper Feather River watershed for water years 1991-2020. Map (a) was created using monthly data summarized from Parameter-elevation Regressions on Independent Slopes Model (PRISM; Eidsenink et al., 2007) which were downscaled to 30-meter grids while map (b) was created using data calculated by the Basin Characterization Model (Flint et al., 2021) in millimeters per year (mm/yr).

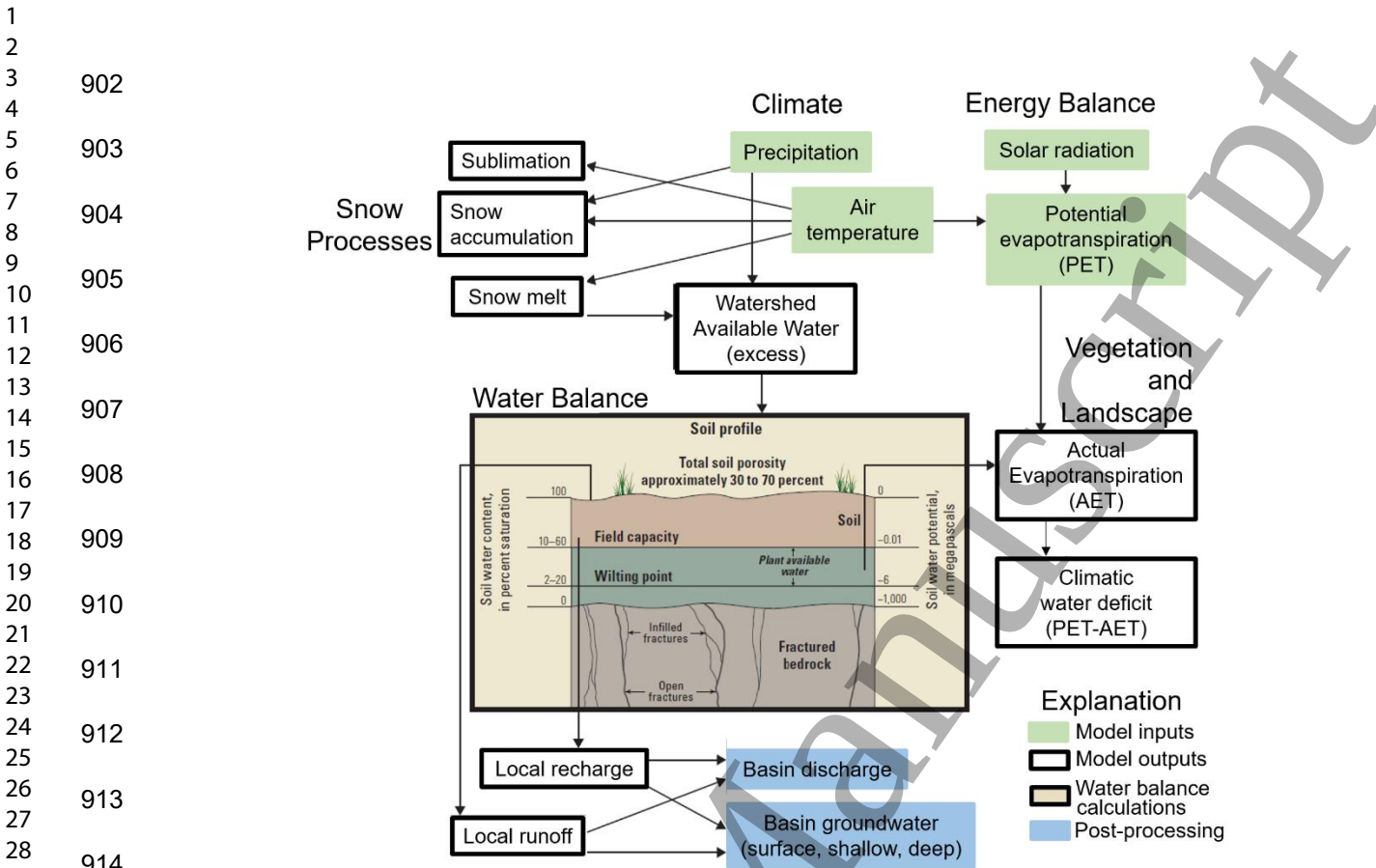
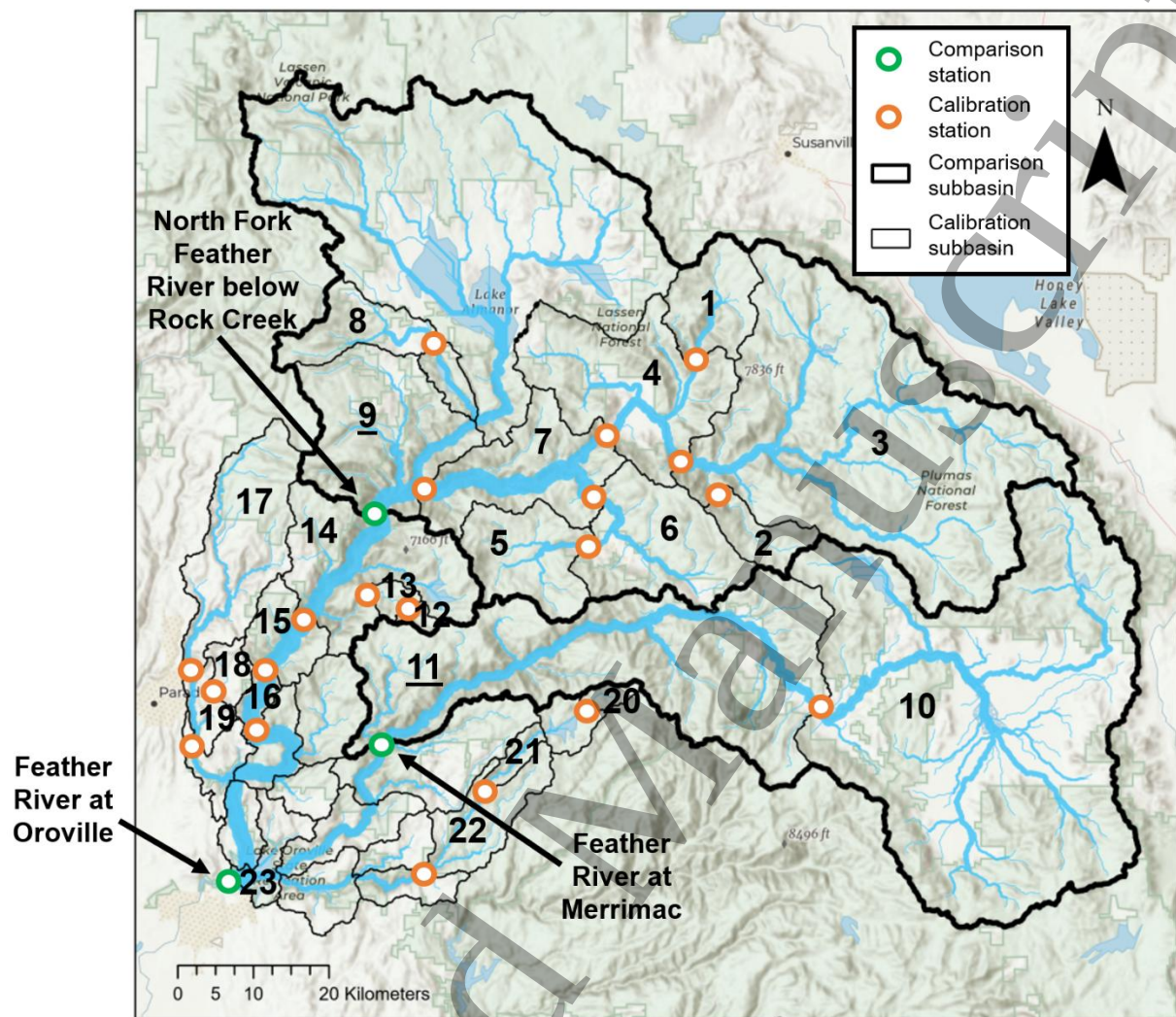


Figure 3: Schematic of the inputs and outputs and major water balance components within the Basin Characterization Model modified from Flint et al. (2021).

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Figure 4: Upper Feather River watershed with streamflow calibration stations (orange circles) and their associated subbasins (thin black lines) used for model calibration (Table 1), and the three comparison stations (green dots; underlined numbers, Table 1) and their upstream subbasins (thick black lines) used to quantify changes in actual evapotranspiration, soil moisture, and streamflow.

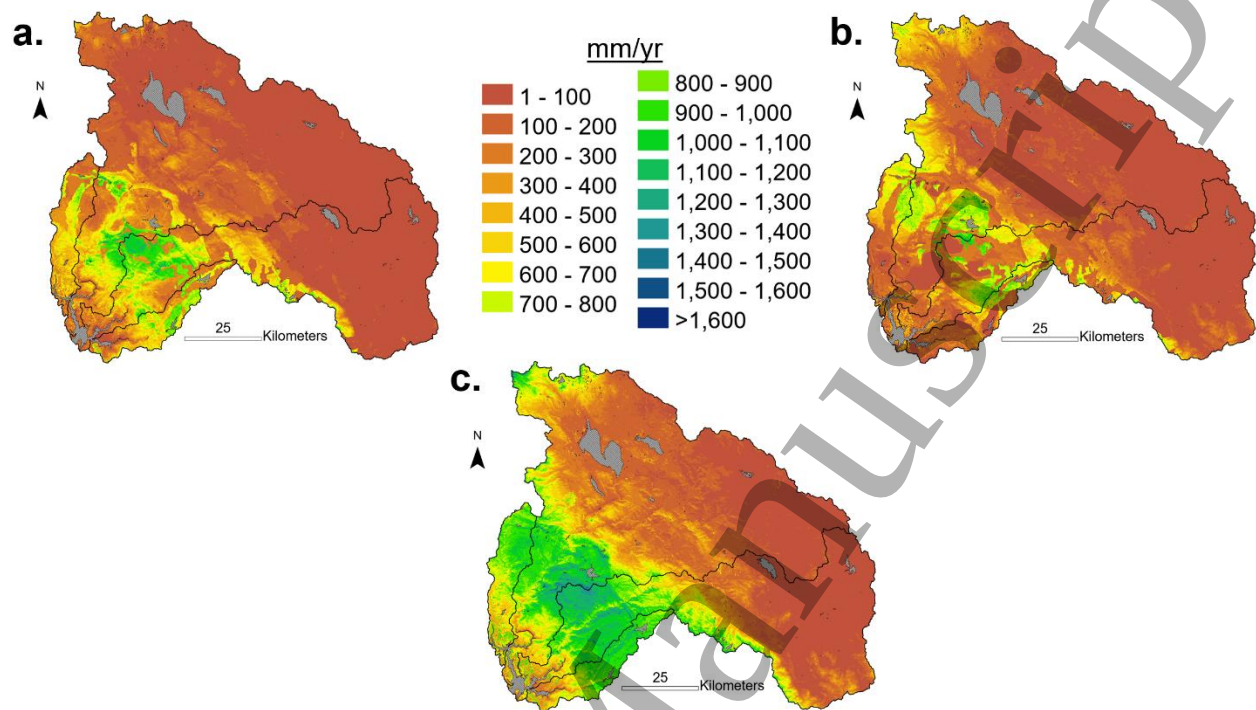


Figure 5: Average annual (a) runoff, (b) recharge, and (c) runoff plus recharge (water supply) modeled by the Basin Characterization Model (Flint et al., 2021) for the upper Feather River watershed for water years 1991-2020.

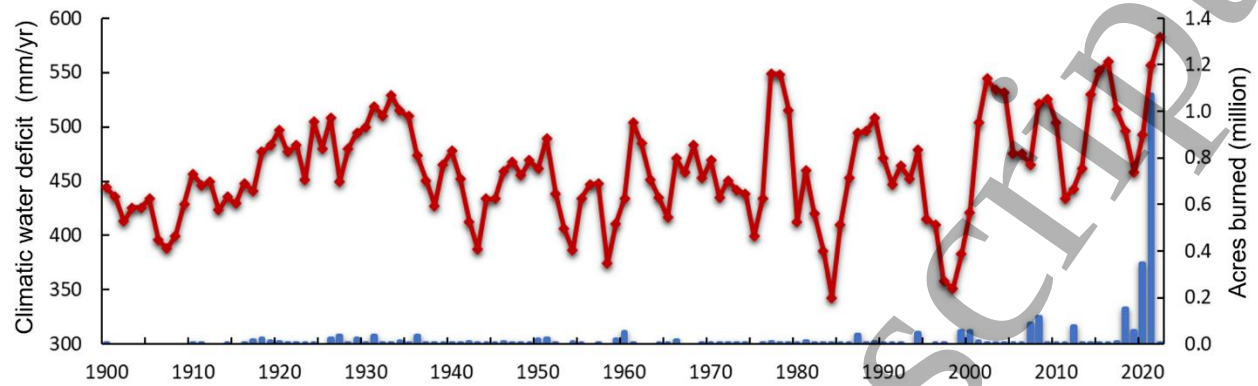


Figure 6: Three-year running average of climatic water deficit from 1900 to 2023 for the upper Feather River watershed (red line) calculated by the Basin Characterization Model. Blue bars indicate the total acres burned in the watershed by calendar year from the California Department of Forest and Fire Protection's Fire and Resource Assessment Program (FRAP) dataset (CALFIRE, 2015).

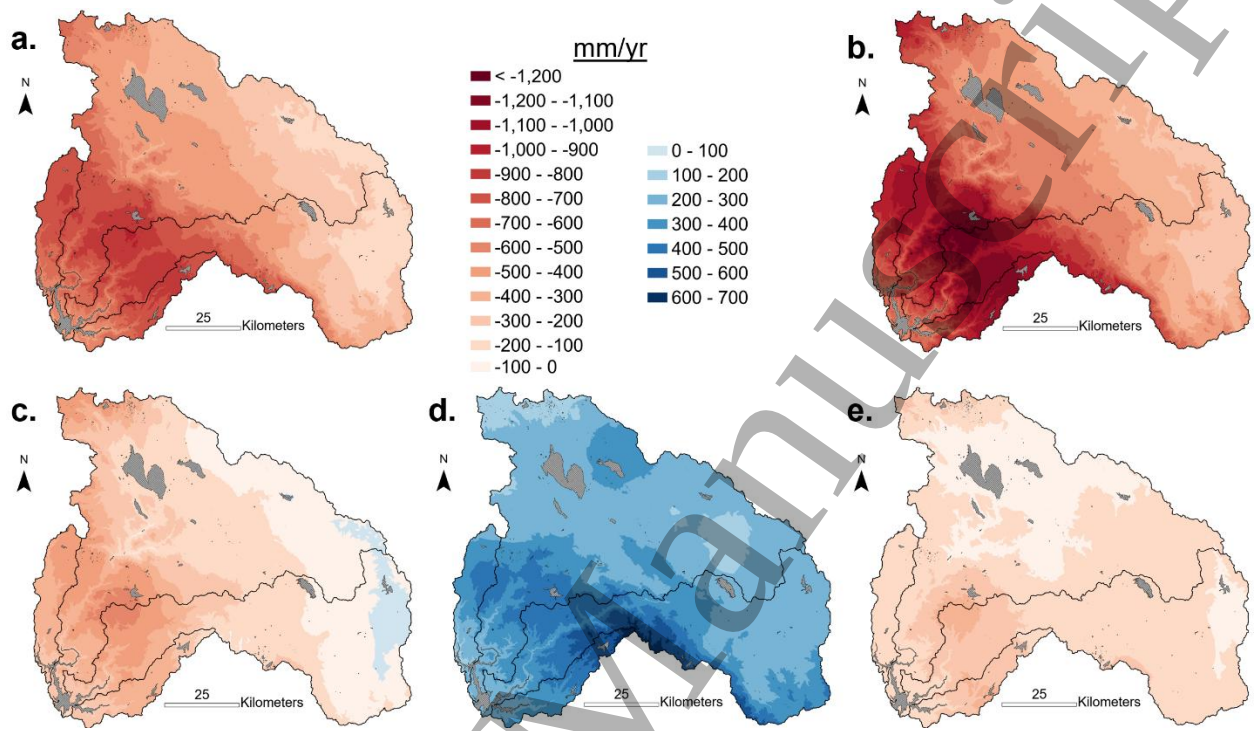


Figure 7: Average annual precipitation (1991-2020) subtracted from annual precipitation in water year (a) 2020, (b) 2021, (c) 2022, (d) 2023, and (e) 2024 in millimeters per year for the upper Feather River watershed. These spatial maps were created using Parameter-elevation Regressions on Independent Slopes Model (PRISM; Eidsenink et al., 2007) data downscaled to 30-meter grids.

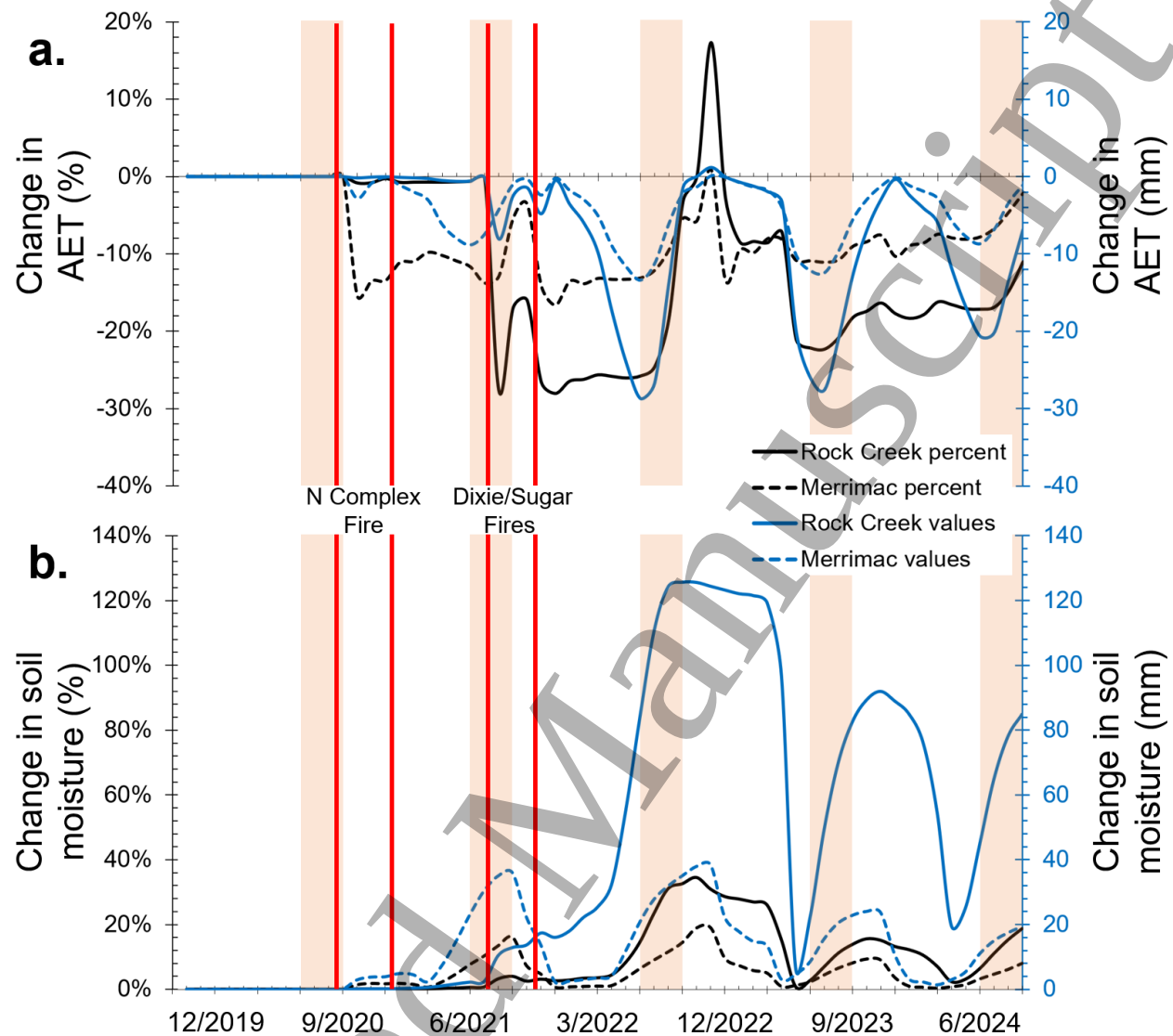


Figure 8: Simulated change in (a) actual evapotranspiration (AET) and (b) soil moisture between postfire and baseline model runs for Feather River at Merrimac (Merrimac on legend; dotted lines) and North Fork Feather River below Rock Creek (Rock Creek on legend; solid lines) subbasins. Relative change (difference between postfire and baseline with respect to baseline, in percent) in black lines and magnitude change (difference between postfire and baseline, in millimeters or mm) in blue lines. The start and end of North Complex and Dixie and Sugar Fires are shown in vertical solid red lines; light orange shaded vertical rectangles denote June-August.

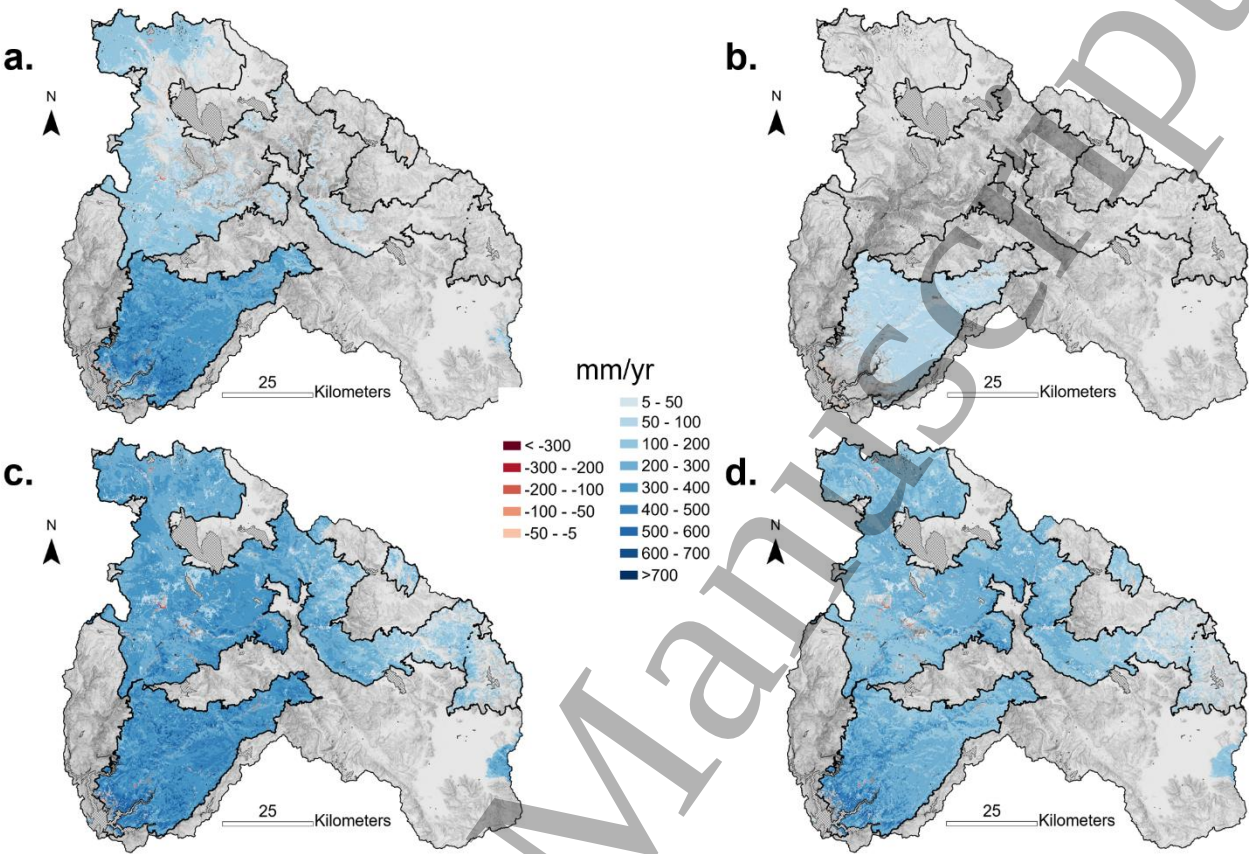


Figure 9: Average annual runoff plus recharge (water supply; 1991-2020) subtracted from annual precipitation in for (a) 2021, (b) 2022, (c) 2023, and (d) 2024 in millimeters per year for the upper Feather River watershed. Recharge and runoff modeled by the Basin Characterization Model (Flint et al., 2021).

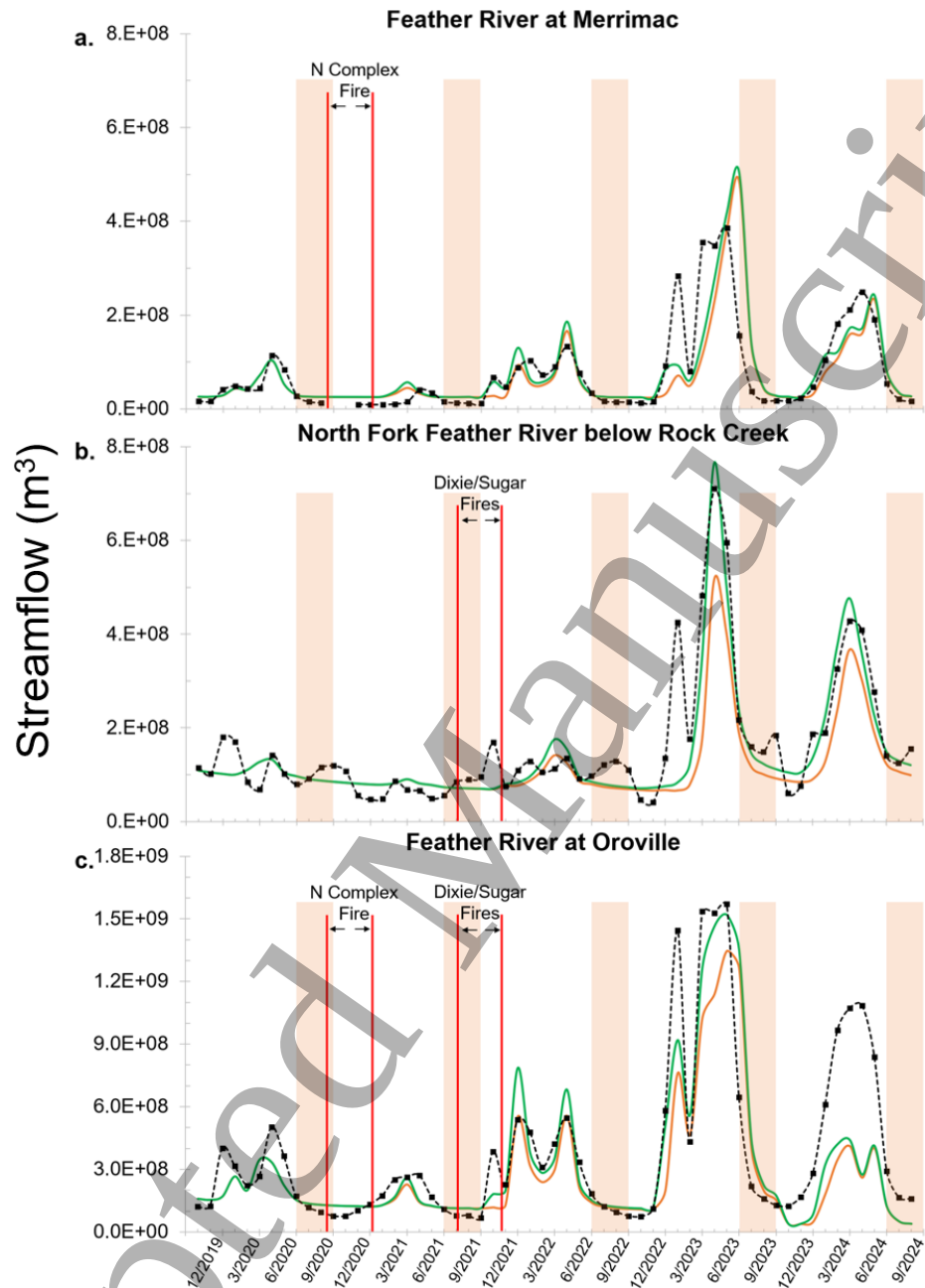


Figure 10: Simulated streamflow for baseline (orange line) and postfire (green line) for (a) Feather River at Merrimac, (b) North Fork Feather River below Rock Creek and (c) Feather River at Oroville. Start and end of North Complex and Dixie/Sugar Fires are shown in vertical solid red lines; light orange shaded vertical rectangles denote June-August. Monthly observed or reference data (see Table 1 for details) are shown in black dotted lines.

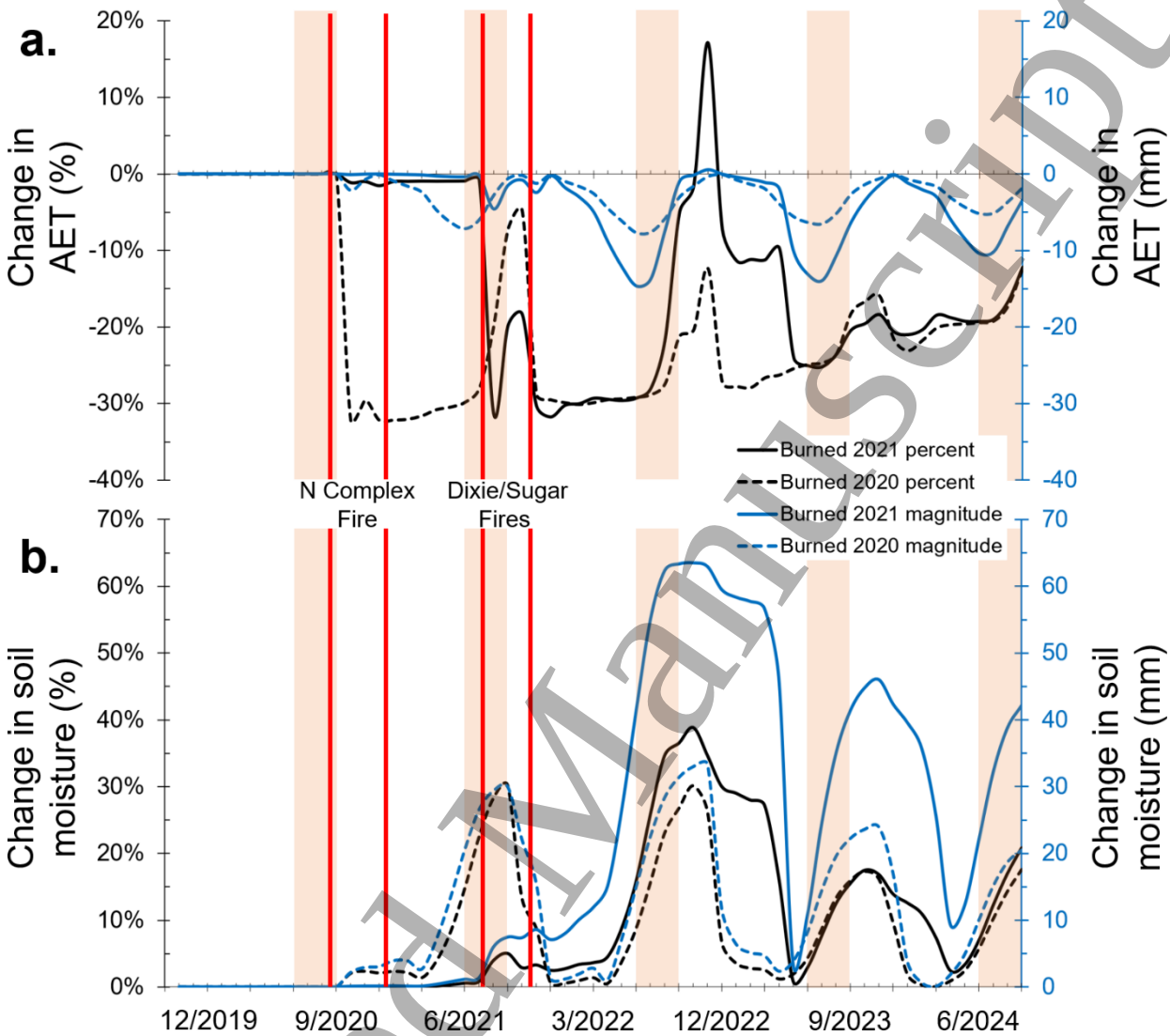


Figure 11: Simulated changes in (a) actual evapotranspiration and (b) soil moisture storage between postfire and baseline for burned 2020 (dashed) and burned 2021 (solid) areas within the upper Feather River watershed (see Tables S3 and S4). Relative change (difference between postfire and baseline with respect to baseline, in percent) in black lines and magnitude change (difference between postfire and baseline, in millimeters or mm) in blue lines. Start and end of North Complex and Dixie/Sugar Fires shown in vertical solid red lines; light orange shaded vertical rectangles denote June-August.

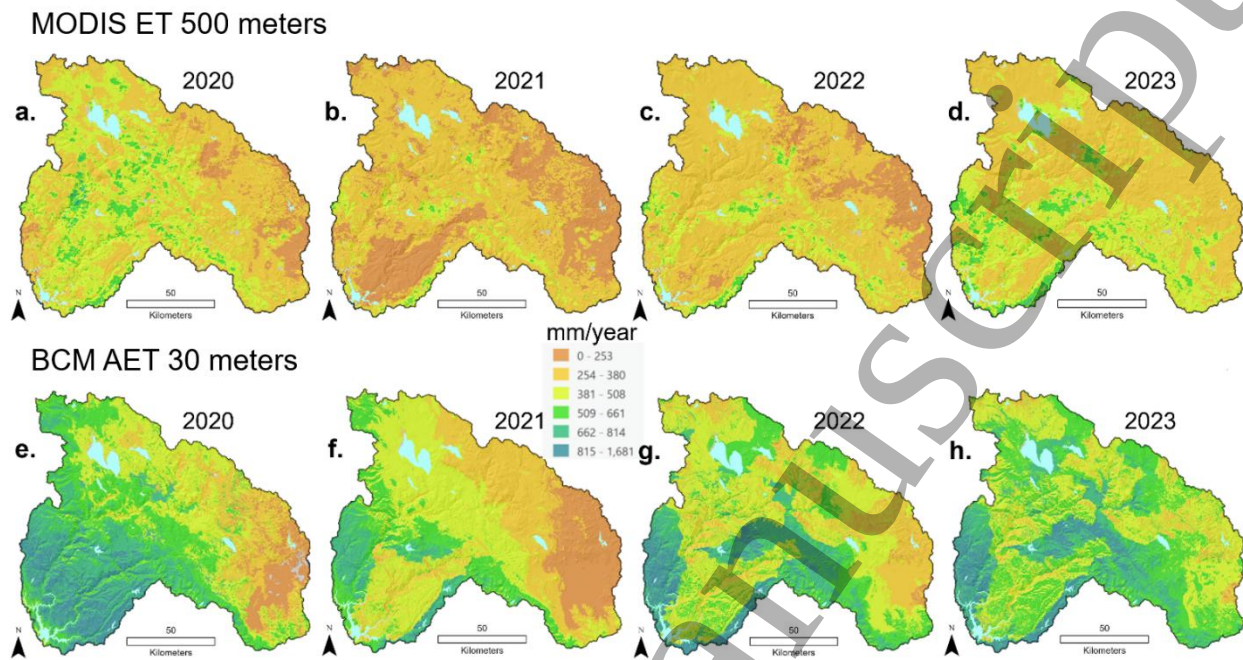


Figure 12: Calendar year total (a-d) MODIS evapotranspiration (ET; Running et al., 2021) and (e-h) Basin Characterization Model (Flint et al., 2021) simulated actual evapotranspiration (AET) in mm per year.

Table 1: Upper Feather River watershed calibration stations from the U.S. Geological Survey (USGS) and California Department of Water Resources (DWR) and associated subbasins used in this study (see Fig. 4 for location). Stations used for comparison are bold and have a rectangular border. Data summarized from USGS, 2024 and CDEC, 2025.

Subbasin	Agency	Name of Station	Station ID	Diversi on Station	Upstream Area (km ²)	Period of Record used
1	USGS	Lights Creek near Taylorsville	11401300	-	148	10/1957-09/1962
2	USGS	Little Grizzly Creek near Negesee	11401180	-	78	09/1964-09/1979
3	USGS	Indian Creek near Taylorsville	11401200	-	1366	10/1957-09/1980
4	USGS	Indian Creek near Crescent Mills	11401500	-	1913	10/1930-09/1993
5	USGS	Spanish Creek above Quincy	11401920	-	194	12/2009-09/2024
6	USGS	Spanish Creek above Blackhawk Creek at Keddie	11402000	-	476	10/1933-09/2024
7	USGS	East Branch North Fork Feather River near Rich Bar	11403000	-	2658	12/1967-09/1982
8	USGS	Butt Creek above Almanor	11400000	-	127	10/1936-09/1964
9	USGS	North Fork Feather River below Rock Creek	11403200	11403800	3193	10/1986-09/2024
10	USGS	Middle Fork Feather River near Clio	11392500	-	1777	10/1925-09/1979
11	DWR	Feather River at Merrimac	MER	-	2750	11/1986-09/2024
12	USGS	Grizzly Creek near Storrie	11404000	-	14	10/1929-09/1944
13	USGS	Grizzly Creek below Diversion Dam	11404300	11403700	37	10/1985-09/2024
14	USGS	North Fork Feather River below Grizzly Creek	11404330	11404360	372	10/1986-09/2000
15	USGS	North Fork Feather River at Pulga	11404500	-	468	04/1911-05/1958
16	USGS	North Fork Feather River above Big Bend	11405000	-	499	01/1906-12/1910
17	USGS	West Branch Feather River near Paradise	11405300	-	291	10/1957-09/1986
18	USGS	Concow Creek near Yankee Hill	11406000	-	39	10/1927-09/1952
19	USGS	West Feather River near Yankee Hill	11406500	-	377	10/1930-09/1963
20	USGS	South Fork Feather River above Little Grass Valley Reservoir	11394800	-	21	10/1960-09/1979
21	USGS	South Fork Tunnel near Strawberry Valley	11395150	11395200	97	10/1973-09/2006
22	USGS	South Fork Feather River below Forbestown Dam	11396200	11396290	227	10/1972-09/2010
23	DWR	Feather River at Oroville	FTO	-	9334	10/1905-09/2024

Table 2: Simulated annual postfire-to-baseline recharge plus runoff response in cubic meters (m³) and acre feet (AF) within the upper Feather River watershed boundary.

	2021	2022	2023	2024
	m ³ (AF)	m ³ (AF)	m ³ (AF)	m ³ (AF)
Baseline	2.3E+08 (186,827)	1.5E+09 (1,215,393)	4.6E+09 (3,720,502)	3.2E+09 (2,557,402)
Postfire	2.9E+08 (232,889)	2.0E+09 (1,649,686)	5.8E+09 (4,729,809)	4.0E+09 (3,247,522)
Difference	5.7E+07 (46,062)	5.4E+08 (434,293)	1.2E+09 (1,009,306)	8.5E+08 (690,119)
Percent Difference	24.7%	35.7%	27.1%	27.0%
Change relative to average year	1.6%	14.8%	34.3%	23.5%

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1048 Table 3: Model statistics for baseline and postfire simulations including coefficient of determination (R^2),
1049 Nash-Sutcliffe efficiency (NSE) and percent bias (PBIAS). For locations see Figure 4 and station details
1050 see Table 1.

	Feather River at Merrimac				North Fork Feather River below Rock Creek				Feather River at Oroville			
	Calibrated Baseline	Baseline (2020-2024)	Postfire (2020-2024)	Difference	Calibrated Baseline	Baseline (2020-2024)	Postfire (2020-2024)	Difference	Calibrated Baseline	Baseline (2020-2024)	Postfire (2020-2024)	Difference
Measured Flow (m ³)	-	4.33E+09	-	-	-	9.46E+09	-	-	-	2.26E+10	-	-
BCM flow (m ³)	-	4.08E+09	4.56E+09	12.0%	-	7.02E+09	8.46E+09	20.5%	-	1.60E+10	1.86E+10	16.1%
Monthly R ²	0.64	0.50	0.56	0.06	0.57	0.72	0.79	0.07	0.81	0.66	0.70	0.04
NSE	0.62	0.45	0.50	0.05	0.51	0.59	0.77	0.18	0.81	0.59	0.68	0.09
PBIAS	0.0%	7.0%	4.3%	-2.7%	0.0%	25.8%	10.5%	-15.2%	0.0%	29.1%	17.7%	-11.4%

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1053 Table 4: Simulated annual postfire-to-baseline watershed response in comparison streamflow stations. The
 1054 runoff coefficient (discharge/precipitation) for each station is also shown. For station details see Figure 4
 1055 and Table 1.

Feather River at Merrimac					North Fork Feather River below Rock Creek				Feather River at Oroville				
		2021	2022	2023	2024	2021	2022	2023	2024	2021	2022	2023	2024
Upstream Area (km ²)		2,750				4,587				9,334			
Annual Precipitation (mm)		433	858	1,327	835	442	790	1,216	849	497	919	1,420	974
Annual Streamflow (m ³)	Baseline	3.2E+08	6.0E+08	1.4E+09	8.5E+08	8.5E+08	1.0E+09	2.1E+09	1.8E+09	1.5E+09	2.6E+09	6.4E+09	2E+09
	Postfire	3.4E+08	7.1E+08	1.6E+09	9.5E+08	8.5E+08	1.1E+09	3.0E+09	2.5E+09	1.5E+09	3.1E+09	7.9E+09	2E+09
Percent Difference		4.9%	18.5%	13.6%	11.5%	0.0%	11.1%	43.6%	35.7%	3.1%	19.0%	22.8%	14.9%
Runoff Coefficient	Baseline	0.27	0.25	0.39	0.37	0.42	0.28	0.38	0.46	0.32	0.31	0.48	0.21
	Postfire	0.29	0.30	0.45	0.41	0.42	0.31	0.54	0.63	0.33	0.37	0.59	0.24